

A convex Bayesian approach to low-rank matrix estimation with heteroscedastic noise

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Abstract

We present a convex Bayesian approach to low-rank matrix estimation under a heteroscedastic noise model. By utilizing the nuclear norm distribution we encourage low-rank structure directly on the data matrix without resorting to a non-convex factorization. To overcome the non-differentiability and non-separability of the prior, we exploit its variational representation using an auxiliary variable. This formulation allows us to develop a Coordinate Ascent Variational Inference (CAVI) algorithm that is strictly convex in all sub-updates, naturally estimates the global regularization parameter, and avoids the Neyman-Scott paradox by explicitly incorporating posterior uncertainty.

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1 Introduction

Low-rank matrix factorization is a workhorse of modern data analysis, underpinning methods ranging from recommender systems to single cell genomics. Yet the overwhelming majority of factorization approaches including non-negative matrix factorization, independent component analysis, and sparse Principal Component Analysis (PCA), optimize a non-convex objective over a product of factor matrices. Non-convexity has practical consequences: different random initializations or minor data perturbations can result in qualitatively different local minima, making results difficult to reproduce and the recovered factor structure ambiguous [1, 2]. Classical PCA is the notable exception, enjoying a globally optimal and unique solution (up to sign flips) via the singular value decomposition, but it cannot handle missing data or heteroskedastic noise, and it provides no principled mechanism for selecting the rank of the solution.

Convex relaxations offer a principled escape from local minima. The nuclear norm, the convex envelope of matrix rank, yields a globally convex objective that implicitly encourages low-rank solutions without factorizing the data matrix [3, 4]. Solving the nuclear-norm-regularized problem produces a unique, reproducible answer regardless of initialization. A practical difficulty, however, is that the strength of the regularization must be chosen by the analyst. Cross-validation (CV) is the standard remedy, but it is computationally expensive and requires arbitrary choices about the CV scheme (i.e. how to choose heldout matrix elements).

We resolve both limitations simultaneously through an empirical Bayes framework. By interpreting the nuclear norm penalty as the negative log-prior of a *nuclear norm* distribution and placing an uninformative Gamma hyperprior on the regularization strength λ , we obtain a fully Bayesian model in which λ is estimated from the data alongside the latent matrix X . Crucially, a naive maximum a posteriori (MAP) approach fails here due to the Neyman–Scott paradox: collapsing posterior uncertainty into a point estimate causes the effective nuclear norm to be underestimated, biasing λ upward and over-shrinking the signal. We sidestep this by developing a Coordinate Ascent Variational Inference (CAVI) algorithm that maintains explicit posterior uncertainty over X , leading to well-calibrated automatic regularization without any cross-validation.

2 Convex Heteroscedastic Matrix Factorization

We consider a setting where we observe a matrix $Y \in \mathbb{R}^{m \times n}$ and a corresponding matrix of known standard errors $S \in \mathbb{R}^{m \times n}$, where s_{ij} is the uncertainty associated with Y_{ij} . The likelihood of the latent true matrix X is elementwise normal,

$$p(Y | X) = \prod_{i=1}^m \prod_{j=1}^n \mathcal{N}(Y_{ij} | X_{ij}, s_{ij}^2) \quad (1)$$

The typical approach would be to represent $X = UV^T$ and optimize over U and V , but this is non-convex and requires careful initialization. In the convex optimization/compressed sensing literature, a common alternative is to place a convex penalty directly on X that encourages low-rank structure. The natural choice is the nuclear norm $\|X\|_* = \sum_i \sigma_i(X)$, the sum of the singular values of X . The loss is then,

$$\mathcal{L}(X) = \sum_{ij} \frac{(Y_{ij} - X_{ij})^2}{s_{ij}^2} + \lambda \|X\|_* \quad (2)$$

In principle λ can be selected by CV, but this is computationally expensive and requires arbitrary choices about the CV scheme (how many folds, how to select heldout entries, what metric to use).

Here, we ask whether Eq 2 can be re-interpreted from an empirical Bayes viewpoint to enable automatic estimation of λ from the data. We consider the nuclear norm penalty as the negative log-prior of a *nuclear norm* distribution, a matrix-variate generalization of the standard Laplace distribution,

$$p(X \mid \lambda) = \frac{1}{Z(\lambda)} \exp(-\lambda \|X\|_*) \quad (3)$$

where the normalizing constant $Z(\lambda) = \text{Vol}(B_*) \cdot (mn)! / \lambda^{mn}$ is the normalizing constant, with $B_* = \{X : \|X\|_* \leq 1\}$ the unit nuclear norm ball.

This distribution encourages low-rank structure in X by penalizing the sum of its singular values, and it is the natural matrix analogue of the Laplace distribution used in Lasso regression for sparsity.

Let $X \in \mathbb{R}^{m \times n}$ be a random matrix. The nuclear norm Prior is a matrix-variate generalization of the standard Laplace (or exponential) distribution, defined by the probability density function:

2.1 Marginal Distribution of the Nuclear Norm

Transforming to generalized polar coordinates where the radius is $R = \|X\|_*$, the volume of the nuclear norm ball grows as R^{mn} . The surface area measure therefore scales as R^{mn-1} . Integrating out the directional components, the marginal density of the nuclear norm itself follows a Gamma distribution,

$$\|X\|_* \sim \text{Gamma}(mn, \lambda). \quad (4)$$

This yields the moments $\mathbb{E}[\|X\|_*] = \frac{mn}{\lambda}$ and $\text{Var}(\|X\|_*) = \frac{mn}{\lambda^2}$.

2.2 The Normalizing Constant

Using the Gamma integral, the normalizing constant $Z(\lambda)$ scales with the volume of the unit nuclear norm ball B_* ,

$$Z(\lambda) = \int_{\mathbb{R}^{m \times n}} \exp(-\lambda \|X\|_*) dX = \text{Vol}(B_*) \frac{(mn)!}{\lambda^{mn}} \quad (5)$$

Thus, $Z(\lambda) \propto \lambda^{-mn}$. This explicit scaling is critical for treating λ as a fully Bayesian random variable.

2.3 Joint Density of Singular Values

Assuming $m \geq n$, the change of variables to the Singular Value Decomposition (SVD) yields the joint density of the non-zero singular values $\sigma_1 > \sigma_2 > \dots > \sigma_n > 0$:

$$p(\sigma_1, \dots, \sigma_n) \propto \prod_{i=1}^n \underbrace{\sigma_i^{m-n} e^{-\lambda \sigma_i}}_{\propto \text{Gamma}(\sigma_i, m-n+1, \lambda)} \cdot \underbrace{\prod_{1 \leq i < j \leq n} (\sigma_i^2 - \sigma_j^2)}_{\text{repulsion term}} \quad (6)$$

This characterizes the singular values as a 1D Coulomb gas subject to a logarithmic pairwise repulsion.

3 Inference

3.1 Coordinate Ascent Variational Inference (CAVI)

To avoid the slow convergence of Markov Chain Monte Carlo (MCMC) while preserving the convexity lost by the common $X = UV^T$ factorization, we utilize a variational representation of the nuclear norm. For any matrix X , the nuclear norm can be expressed as a minimization over an auxiliary symmetric positive definite matrix $Q \in \mathbb{S}_{++}^n$,

$$\|X\|_* = \min_{Q \succ 0} \frac{1}{2} \text{Tr}(XQ^{-1}X^T) + \frac{1}{2} \text{Tr}(Q). \quad (7)$$

Row factorization. Let $q(X) = \prod_{i=1}^m \mathcal{N}(x_i \mid \mu_i, \Sigma_i)$, where $x_i \in \mathbb{R}^n$ is the i -th row of X . The expected second moment is

$$\mathbb{E}_q[X^T X] = \sum_{i=1}^m (\mu_i \mu_i^T + \Sigma_i) \in \mathbb{S}_{++}^n, \quad (8)$$

which pairs directly with the $Q \in \mathbb{S}_{++}^n$ auxiliary variable above. When optimizing the Evidence Lower Bound (ELBO), we must evaluate $\mathbb{E}_q[-\lambda \|X\|_*]$. Substituting the variational representation and applying Jensen's inequality gives the tractable lower bound:

$$\mathbb{E}_q[-\lambda \|X\|_*] \geq -\frac{1}{2} \mathbb{E}_q[\lambda] \text{Tr}(\mathbb{E}_q[X^T X] Q^{-1}) - \frac{1}{2} \mathbb{E}_q[\lambda] \text{Tr}(Q). \quad (9)$$

This requires $O(n^3)$ matrix operations per iteration.

Column factorization. We could in principle factorize $q(X)$ over columns instead of rows, but this is equivalent to the row factorization by simply transposing the input data.

We place a Gamma hyperprior on the regularization parameter: $\lambda \sim \text{Gamma}(a_0, b_0)$. We iteratively update the variational distributions to maximize the ELBO.

Update for Auxiliary Matrix Q . Let $\Psi = \mathbb{E}_q[X^T X] = \sum_{i=1}^m (\mu_i \mu_i^T + \Sigma_i)$. The value of Q that tightly bounds the expectation is the matrix square root of Ψ :

$$Q^* = \Psi^{1/2} = \left(\sum_{i=1}^m \mu_i \mu_i^T + \Sigma_i \right)^{1/2} \quad (10)$$

Update for Variational Posterior $q(X)$. Given Q and $\bar{\lambda} = \mathbb{E}_q[\lambda]$, the log-posterior for each row x_i is a multivariate Gaussian. Let $S_i = \text{diag}(s_{i1}^2, \dots, s_{in}^2)$ be the known noise variance matrix for row i . The optimal variational updates are:

$$\Sigma_i = (S_i^{-1} + \bar{\lambda} Q^{-1})^{-1} \quad (11)$$

$$\mu_i = \Sigma_i (S_i^{-1} y_i) \quad (12)$$

Notice how the auxiliary matrix Q induces correlation across the columns of X , even if the noise S_i is diagonal.

Batched implementation. For settings where n is moderate (e.g. $n \lesssim 300$) but m is large, the bottleneck of full CAVI is the simultaneous storage of all m row covariances $\Sigma_i \in \mathbb{S}_{++}^n$. We address this with `matlap_batched`, which processes rows in mini-batches of size B . Each batch computes Σ_i via Cholesky, immediately extracts $\text{diag}(\Sigma_i)$ (needed for the ELBO) and the batch’s contribution to $\Psi = \sum_i (\mu_i \mu_i^\top + \Sigma_i)$, then discards the full Σ_i before moving to the next batch. Peak memory is $O(Bn^2 + mn)$ rather than $O(mn^2)$. With $B = 64$ and $n = 1,000$ this is approximately 256 MB versus 40 GB. The algorithm is otherwise identical to full CAVI (Section 3.1) and converges to the same solution.

Update for Hyperparameter Posterior $q(\lambda)$. Because $Z(\lambda) \propto \lambda^{-mn}$, the log-prior contributes $mn \log \lambda$. Combined with the Gamma prior and the variational bound on the nuclear norm, the optimal $q(\lambda)$ is $\text{Gamma}(a_N, b_N)$. Notably, the scale update simplifies beautifully because $\text{Tr}(\Psi Q^{-1}) + \text{Tr}(Q) = \text{Tr}(\Psi \Psi^{-1/2}) + \text{Tr}(\Psi^{1/2}) = 2\text{Tr}(\Psi^{1/2})$.

$$a_N = a_0 + mn \quad (13)$$

$$b_N = b_0 + \text{Tr}(\Psi^{1/2}) \quad (14)$$

The expected value is updated as $\bar{\lambda} = \frac{a_N}{b_N}$.

Evidence Lower Bound. Convergence is monitored via the ELBO. Using the variational bound at the optimal $Q = \Psi^{1/2}$ and the normalizing constant $\log Z(\lambda) = -mn \log \lambda + \text{const}$, the ELBO decomposes as

$$\begin{aligned} \mathcal{L} = & \underbrace{-\frac{1}{2} \sum_{(i,j) \in \mathcal{O}} \left[\frac{(y_{ij} - \mu_{ij})^2 + \Sigma_i[j,j]}{s_{ij}^2} + \log(2\pi s_{ij}^2) \right]}_{\text{log-likelihood}} \underbrace{-\bar{\lambda} \text{Tr}(Q) + mn \mathbb{E}_q[\log \lambda]}_{\text{prior on } X \text{ (lower bound)}} \\ & + \underbrace{\frac{1}{2} \sum_{i=1}^m [\log |\Sigma_i| + n(1 + \log 2\pi)]}_{\text{entropy of } q(X)} - \underbrace{\text{KL}(q(\lambda) \parallel p(\lambda))}_{\text{KL for } \lambda} \end{aligned} \quad (15)$$

where $\mathcal{O}$ is the set of observed entries and

$$\text{KL}(\text{Gamma}(a_N, b_N) \parallel \text{Gamma}(a_0, b_0)) = a_N \log b_N - a_0 \log b_0 - \log \Gamma(a_N) + \log \Gamma(a_0) + (a_N - a_0) \psi(a_N) + (16)$$

The full procedure is summarized in Algorithm 1.

Algorithm 1 CAVI for Bayesian Low-Rank Matrix Denoising

Require: Data $Y \in \mathbb{R}^{m \times n}$, Variances $S \in \mathbb{R}^{m \times n}$, Prior a_0, b_0 .

- 1: Initialize $\mu_i = y_i$ and $\Sigma_i = I_n$ for $i = 1, \dots, m$.
- 2: Initialize $\bar{\lambda} = \frac{mn}{\|Y\|_*}$.
- 3: **repeat**
- 4: **1. Estimate uncentered expected covariance:**
- 5: $\Psi \leftarrow \sum_{i=1}^m (\mu_i \mu_i^T + \Sigma_i)$
- 6: **2. Update auxiliary matrix:**
- 7: $Q \leftarrow \Psi^{1/2}$
- 8: **3. Update regularization parameter posterior:**
- 9: $a_N \leftarrow a_0 + mn$
- 10: $b_N \leftarrow b_0 + \text{Tr}(Q)$
- 11: $\bar{\lambda} \leftarrow \frac{a_N}{b_N}$
- 12: **4. Update latent matrix posterior row-by-row:**
- 13: **for** $i = 1, \dots, m$ **do**
- 14: $S_i \leftarrow \text{diag}(s_{i1}^2, \dots, s_{in}^2)$
- 15: $\Sigma_i \leftarrow (S_i^{-1} + \bar{\lambda} Q^{-1})^{-1}$
- 16: $\mu_i \leftarrow \Sigma_i S_i^{-1} y_i$
- 17: **end for**
- 18: **until** ELBO converges

Ensure: Posterior mean $M = [\mu_1, \dots, \mu_m]^T$, uncertainties $\{\Sigma_i\}_{i=1}^m$, expected penalty $\bar{\lambda}$.

Missing Data. The model naturally accommodates missing observations. If entry (i, j) is unobserved, we set $s_{ij} = \infty$, making the noise precision $1/s_{ij}^2 = 0$. The entry then contributes nothing to the likelihood, the row posterior update for μ_i , or the ELBO. Observed and missing entries can be arbitrarily interleaved.

Rényi α -ELBO for λ selection. The standard ELBO lower-bounds $\log p(Y | \lambda)$ but is tight only when $q(X) = p(X | Y, \lambda)$. For the purpose of selecting λ , we exploit the Rényi α -ELBO [5],

$$\mathcal{L}_\alpha = \frac{1}{1-\alpha} \log \int q(X)^\alpha p(Y, X | \lambda)^{1-\alpha} dX, \quad 0 \leq \alpha < 1, \quad (17)$$

which satisfies $\mathcal{L}_\alpha \leq \log p(Y | \lambda)$ and recovers the ELBO as $\alpha \rightarrow 1$; at $\alpha = 0$ it is the importance-weighted objective.

For the low-rank-plus-isotropic model with diagonal posterior $q(x_{ij}) = \mathcal{N}(\mu_{ij}, \sigma_{ij}^2)$, Gaussian likelihood $p(y_{ij} | x_{ij}) = \mathcal{N}(y_{ij}, s_{ij}^2)$, and prior marginal $p(x_{ij}) = \mathcal{N}(0, v_j(\lambda))$, the Rényi integral is a product of Gaussian factors and evaluates in closed form as a correction to the diagonal-Gaussian ELBO:

$$\mathcal{L}_\alpha = \mathcal{L} + \frac{1}{\beta} \sum_{(i,j) \in \mathcal{O}} \left[\frac{\beta^2 f_{ij}^2}{2(1-2\beta g_{ij})} - \frac{1}{2} \log(1-2\beta g_{ij}) \right], \quad (18)$$

where $\beta = 1 - \alpha$,

$$f_{ij} = \sigma_{ij} \left(\frac{y_{ij} - \mu_{ij}}{s_{ij}^2} - \frac{\mu_{ij}}{v_j(\lambda)} \right), \quad (19)$$

$$g_{ij} = \frac{1}{2} \left(1 - \frac{\sigma_{ij}^2}{s_{ij}^2} - \frac{\sigma_{ij}^2}{v_j(\lambda)} \right), \quad (20)$$

and the sum is over observed entries $\mathcal{O}$. The correction vanishes as $\alpha \rightarrow 1$ (recovering $\mathcal{L}$) and is negative for $\alpha \in (0, 1)$ (a tighter bound). Crucially, $v_j(\lambda)$ depends on λ through the low-rank-plus-isotropic parameterization, so $\mathcal{L}_\alpha$ is differentiable with respect to λ and can be maximized by gradient-based search (e.g. BFGS on $\log \lambda$). In our implementation (`matlap_iso_renyi_lambda`) we alternate between CAVI updates for the posterior q at fixed λ and a one-dimensional BFGS step to update λ by maximizing (18), providing a fully automatic, cross-validation-free procedure for joint posterior and regularization inference.

3.2 Low-Rank CAVI via the Woodbury Identity

The full CAVI algorithm stores the m row covariance matrices $\{\Sigma_i\}$ explicitly, requiring $O(mn^2)$ memory—approximately 40 GB for a $10,000 \times 1,000$ matrix in 32-bit precision. We describe two families of extensions that reduce this to $O(mnr)$ or $O(mn)$ while preserving the key algorithmic properties.

We restrict the variational family to a rank- r *factor subspace*. Instead of maintaining free row covariances $\Sigma_i \in \mathbb{S}_{++}^n$, we fix a shared loading matrix $V_r \in \mathbb{R}^{n \times r}$ with orthonormal columns and parameterise $x_i = V_r z_i$ where $z_i \in \mathbb{R}^r$. The variational posterior is

$$q(z_i) = \mathcal{N}(z_i \mid \hat{z}_i, A_r^{(i)-1}), \quad x_i = V_r z_i, \quad (21)$$

so that the implied row posterior is $\mathcal{N}(x_i \mid \mu_i, \Sigma_i)$ with $\mu_i = V_r \hat{z}_i$ and $\Sigma_i = V_r A_r^{(i)-1} V_r^\top$.

Per-row update. Let $p_i \in \mathbb{R}^n$ be the vector of observed precisions for row i (i.e. $p_{ij} = 1/s_{ij}^2$ for observed entries, 0 for missing). The $r \times r$ factor precision and factor mean are:

$$A_r^{(i)} = \text{diag}(\bar{\lambda}/d_r) + V_r^\top \text{diag}(p_i) V_r, \quad (22)$$

$$\hat{z}_i = A_r^{(i)-1} \left(V_r^\top (p_i \odot y_i) \right), \quad (23)$$

where d_r are the eigenvalues from the current Q update (so $Q^{-1} = V_r \text{diag}(1/d_r) V_r^\top$). Each $A_r^{(i)}$ is $r \times r$, so the Cholesky solve costs $O(nr + r^3)$ per row and $O(mnr)$ over all rows—an n/r -fold reduction from the full $O(mn^3)$.

Q update in factor space. We accumulate the expected second moment restricted to the factor subspace:

$$\Psi_r = \sum_{i=1}^m \left(\hat{z}_i \hat{z}_i^\top + A_r^{(i)-1} \right) \in \mathbb{S}_{++}^r. \quad (24)$$

The eigendecomposition $\Psi_r = W_r \text{diag}(d_r) W_r^\top$ gives the matrix square root $Q_r = W_r \text{diag}(\sqrt{d_r}) W_r^\top$, and the loading matrix is rotated: $V_r \leftarrow V_r W_r$. With this convention the Q update diagonal is d_r , $\text{Tr}(Q) = \sum_k \sqrt{d_{r,k}}$, and the ELBO entropy for row i is $\frac{1}{2} \log |A_r^{(i)-1}| = -\frac{1}{2} \log |A_r^{(i)}|$.

Initialization. V_r is initialized via a randomized SVD (see Section 6.1) of the observed data matrix, projecting Y onto its top- r right singular vectors.

Memory and complexity. The algorithm stores $\hat{z}_i \in \mathbb{R}^r$ and $A_r^{(i)} \in \mathbb{R}^{r \times r}$ per row, giving $O(mr^2 + nr)$ storage instead of $O(mn^2)$. At $m = 10,000$, $n = 1,000$, $r = 50$ this is approximately 44 MB versus 40 GB.

ELBO in factor space. Because x_i lives in the column space of V_r , the entropy of $q(x_i)$ is computed in the r -dimensional factor space: $H[q(z_i)] = \frac{1}{2} \log |A_r^{(i)-1}| + \frac{r}{2} (1 + \log 2\pi)$. The λ update uses mn replaced by mr in the Gamma shape parameter update.

3.3 Low-Rank-Plus-Isotropic CAVI via the Woodbury Identity

The factor-space CAVI restricts Q to be rank- r , assigning zero prior precision to off-subspace directions and forcing all posterior mass onto $\text{col}(V_r)$. We instead restrict Q to a low-rank-plus-isotropic family,

$$Q = V_r \text{diag}(d_r) V_r^\top + \delta (I - V_r V_r^\top), \quad (25)$$

where $\delta > 0$ is an additional *variational* parameter—it has the same status as the in-subspace eigenvalues $d_{r,k}$, representing the off-subspace degrees of freedom of Q and optimized by CAVI alongside them.

Update for δ . With $Q^{-1} = V_r \text{diag}(1/d_r) V_r^\top + \delta^{-1}(I - V_r V_r^\top)$, the nuclear-norm ELBO bound decomposes into independent in-subspace and off-subspace terms. The off-subspace contribution $-\frac{\bar{\lambda}}{2} [\text{Tr}(\Psi_\perp)/\delta + (n-r)\delta]$ is minimised in closed form:

$$\delta^* = \sqrt{\frac{\text{Tr}(\Psi_\perp)}{n-r}}, \quad \text{Tr}(\Psi_\perp) = \text{Tr}(\Psi) - \text{Tr}(\Psi_r), \quad (26)$$

where $\text{Tr}(\Psi) = \sum_i (\|\mu_i\|^2 + \text{Tr}(\Sigma_i))$ and $\text{Tr}(\Psi_r) = \sum_k d_{r,k}^2$. This is the exact analogue of $d_{r,k}^* = \sqrt{[\Psi_r]_{kk}}$ for the in-subspace modes: δ^* is the RMS off-subspace eigenvalue of Ψ .

Posterior updates. Let $\gamma = \bar{\lambda}/\delta$ (off-subspace prior precision), $\tilde{p}_{ij} = p_{ij} + \gamma$, and $c_k = \bar{\lambda}/d_{r,k} - \gamma$. In practice $\delta < d_{r,k}$ (off-subspace modes of Q are smaller than in-subspace), so $c_k < 0$; the posterior precision $\Lambda_i = \text{diag}(\tilde{p}_i) + V_r \text{diag}(c_k) V_r^\top$ is nevertheless PD. The Woodbury identity gives

$$\Sigma_i = \text{diag}(\tilde{p}_i^{-1}) - \text{diag}(\tilde{p}_i^{-1}) V_r \tilde{B}_i^{-1} V_r^\top \text{diag}(\tilde{p}_i^{-1}), \quad (27)$$

$$\tilde{B}_i = \text{diag}(1/c_k) + V_r^\top \text{diag}(\tilde{p}_i^{-1}) V_r \in \mathbb{R}^{r \times r}, \quad (28)$$

where $\tilde{B}_i$ is symmetric and invertible (since $\Lambda_i \succ 0$ implies $\det(\tilde{B}_i) \neq 0$ via the matrix determinant lemma), but need not be positive definite when $c_k < 0$; in practice we diagonalise $\tilde{B}_i = E \text{diag}(\ell) E^\top$ via `eigh` rather than Cholesky. The posterior mean $\mu_i = \Sigma_i \text{diag}(p_i) y_i$ follows the same structure and has components outside $\text{col}(V_r)$, capturing signal not explained by the current factor subspace. $\text{diag}(\Sigma_i)$ is extracted in $O(nr)$ without forming Σ_i explicitly.

Q and δ updates. $\text{Tr}(\Psi_r)$ and V_r are updated exactly as in Section 3.2. $\text{Tr}(\Psi)$ is accumulated in $O(mn)$ from $\|\mu_i\|^2$ and $\text{diag}(\Sigma_i)$, giving $\text{Tr}(\Psi_\perp)$ and thus $\delta^* = \sqrt{\text{Tr}(\Psi_\perp)/(n-r)}$ and $\gamma \leftarrow \bar{\lambda}/\delta^*$ in $O(n)$. $\text{Tr}(Q) = \sum_k d_{r,k} + (n-r)\delta^*$.

ELBO. By the matrix determinant lemma applied to $\Lambda_i = \text{diag}(\tilde{p}_i) + V_r \text{diag}(c_k) V_r^\top$,

$$\log |\Sigma_i| = - \sum_{k=1}^r \log |c_k| - \log |\tilde{B}_i| - \sum_{j=1}^n \log(p_{ij} + \gamma), \quad (29)$$

an $O(n + r^3)$ computation that accounts for the full n -dimensional entropy. Absolute values are required because $c_k < 0$ in the typical case ($\delta < d_{r,k}$); since $\Lambda_i \succ 0$, the signs of $\prod_k c_k$ and $\det(\tilde{B}_i)$ agree, so $|\det(\Sigma_i)| = (\prod_j \tilde{p}_j)^{-1} \cdot |\prod_k c_k|^{-1} \cdot |\det(\tilde{B}_i)|^{-1}$ is real and positive.

Summary. At the same $O(mnr + mr^3)$ per-iteration cost and $O(mr^2 + mn)$ memory as the factor-space CAVI: (i) μ_i captures off-subspace signal; (ii) the entropy is n -dimensional, correcting the auto- λ over-shrinkage; (iii) no new hyperparameters are introduced— δ is optimized from the data in closed form each iteration.

4 Theoretical Properties

4.1 Invariance of the Optimal Regularisation to Observation Density

A practically important consequence of the Bayesian formulation is that the oracle-optimal regularisation strength is approximately invariant to the fraction of missing observations, while the analogous proximal estimator is not.

Setup. Let $X_{\text{true}} \in \mathbb{R}^{m \times n}$ be a rank- r matrix. Each entry is independently observed with probability f ; observed entries carry heteroscedastic Gaussian noise $y_{ij} = x_{ij} + \varepsilon_{ij}$, $\varepsilon_{ij} \sim \mathcal{N}(0, s_{ij}^2)$. We use the noise-calibrated heuristic

$$\bar{\lambda} = \frac{\sqrt{\max(m, n)}}{\sqrt{\text{median}_{(i,j) \in \Omega}(1/s_{ij}^2)}}, \quad (30)$$

which estimates the typical noise singular-value scale of an $m \times n$ Gaussian matrix with the observed noise distribution (cf. the Marchenko–Pastur bulk edge $\approx (1 + \sqrt{n/m}) s \sqrt{m}$ in the homoscedastic case).

The heuristic is independent of f . For a random Bernoulli(f) mask applied to noise variances $\{s_{ij}^2\}$ drawn iid from any distribution, the observed entries $\{(i, j) \in \Omega\}$ are a uniformly random subsample of all entries. Therefore $\text{median}_{(i,j) \in \Omega}(1/s_{ij}^2)$ converges to the population median by the law of large numbers, and $\bar{\lambda}$ in (30) is independent of f . This holds for arbitrary heteroscedastic noise.

The oracle Bayesian regularisation is also independent of f . In the factor representation, each latent row score $z_i \in \mathbb{R}^r$ has prior $z_{ik} \sim \mathcal{N}(0, d_{r,k}/\lambda)$. The effective data precision for factor k in row i is

$$\beta_k^{(i)} = \sum_j p_{ij} v_{kj}^2 / s_{ij}^2 = f B_k^{(i)}, \quad (31)$$

where $B_k^{(i)} = \sum_j v_{kj}^2 / s_{ij}^2$ is the full-data precision (independent of f). The posterior mean is $\hat{z}_{ik} = \gamma_{ik} z_{ik}^{\text{true}} + \text{noise}$ with shrinkage $\gamma_{ik} = \beta_k^{(i)} / (\beta_k^{(i)} + \lambda / d_{r,k})$. Setting $A = (z_{ik}^{\text{true}})^2$, the MSE for factor (i, k) is

$$\text{MSE}_{ik}(\gamma) = (1 - \gamma)^2 A + \gamma^2 / \beta_k^{(i)}, \quad (32)$$

which is minimised at $\gamma_{ik}^* = A \beta_k^{(i)} / (1 + A \beta_k^{(i)})$. The corresponding optimal regularisation satisfies

$$\frac{\lambda^*}{d_{r,k}} = \beta_k^{(i)} \frac{1 - \gamma_{ik}^*}{\gamma_{ik}^*} = \beta_k^{(i)} \cdot \frac{1}{\beta_k^{(i)} A} = \frac{1}{(z_{ik}^{\text{true}})^2}, \quad (33)$$

since $\beta_k^{(i)}$ cancels exactly. Critically, f (which enters only through $\beta_k^{(i)} = f B_k^{(i)}$) also cancels. The oracle λ^* is therefore independent of both f and the noise heteroscedasticity $\{s_{ij}^2\}$. Since $\bar{\lambda}$ from (30) is also independent of f , we have

$$\frac{\lambda^*}{\bar{\lambda}} = \text{const.}, \quad (34)$$

independent of observation density. Empirically we observe $\lambda^* / \bar{\lambda} \approx 0.74$ across densities from 2% to 90% observed (see Section 7).

Why the proximal estimator is not invariant. For the nuclear-norm penalised least squares problem, the appropriate threshold is the largest singular value of the noise component of Y_Ω . Under a Bernoulli(f) mask, the noise matrix $E_\Omega = \varepsilon \odot P$ has entry variances $f s_{ij}^2$, so its spectral norm scales as $\sqrt{f} \cdot s_{\text{rms}} \sqrt{\max(m, n)}$ (Marchenko–Pastur bulk edge for the sparse noise matrix). The oracle proximal threshold therefore satisfies $\lambda_{\text{prox}}^* \propto \sqrt{f}$, which decreases with f . Relative to the f -independent heuristic (30), the ratio $\lambda_{\text{prox}}^* / \bar{\lambda} \propto \sqrt{f}$ is not constant, so the same grid centre becomes increasingly misspecified as observations become sparse. The Bayesian model avoids this because f enters symmetrically into both the data precision $\beta_k^{(i)}$ and the posterior variance $1/\beta_k^{(i)}$, and these cancel in the bias–variance optimum (33).

4.2 Optimization Landscape and Global Convergence

Although the posterior $p(X \mid Y, S, \lambda, Q)$ is log-concave in X —the Gaussian likelihood contributes a log-quadratic and the GSM prior with Q fixed contributes $-\frac{\bar{\lambda}}{2} \text{Tr}(X^\top Q^{-1} X)$ —the ELBO over the full variational family $(q(X), Q, q(\lambda))$ is not jointly concave. Nonetheless, the CAVI iteration enjoys strong convergence guarantees because every coordinate update has a *unique* global optimum.

Global optimality of `matlap_batched`. Given Q and $\bar{\lambda}$, the optimal $q(x_i)$ is the exact Gaussian posterior $\mathcal{N}(\mu_i, \Sigma_i)$ with $\Sigma_i = (S_i^{-1} + \bar{\lambda}Q^{-1})^{-1}$ and $\mu_i = \Sigma_i S_i^{-1} y_i$, which is unique. The Q -update $Q^* = \Psi^{1/2}$ follows from the AM–GM inequality $\text{Tr}(\Psi Q^{-1}) + \text{Tr}(Q) \geq 2\text{Tr}(\Psi^{1/2})$, with equality at $Q = \Psi^{1/2}$, and $\text{Tr}(\Psi Q^{-1}) + \text{Tr}(Q)$ is strictly convex in $Q \succ 0$ so the minimum is unique. The λ -update is the mean of a Gamma posterior, again unique. Because rows are conditionally independent given (Q, λ) , the mean-field approximation is *exact*: the CAVI iterates converge to the true posterior factorization, which is the unique global maximum of the ELBO. `matlap_batched` is therefore initialization-independent.

Low-rank variants and the frozen-subspace limitation. The low-rank CAVI (`matlap_lowrank`, Section 3.2) initialises a loading matrix V_r from the top- r right singular vectors of Y_Ω and restricts $x_i \in \text{col}(V_r)$ throughout. The Q_r -update rotates V_r *within* its column space but never changes the subspace itself. Within this restriction the CAVI sub-updates are again unique, and the algorithm converges to the global maximum of the *restricted* ELBO. However, if any signal lies outside $\text{col}(V_r^{\text{init}})$ —which is likely when the true rank exceeds r or the signal-to-noise ratio is low—the restricted optimum can be substantially worse than the global one.

The low-rank-plus-isotropic variant (`matlap_lowrank_isotropic`, Section 3.3) partially addresses this: the posterior mean μ_i is full-rank (the isotropic off-subspace precision δ provides gradient signal in all directions), but the covariance structure Q remains restricted to $\text{col}(V_r)$. In practice this gives lower RMSE than the pure low-rank model, though the subspace still does not evolve.

Methods with a free subspace. The Factor Analysis EM (`matlap_faem`) and gradient marginal likelihood (`matlap_gradml`) methods described in Section 6.4 parameterise an unconstrained loading matrix $W_r \in \mathbb{R}^{n \times r}$. The column space of W_r evolves freely across iterations, allowing the model to discover the true signal subspace from data rather than inheriting it from the initialization. The FA EM M-step for W_r is a per-column weighted least-squares problem, globally concave in W_r for fixed posterior moments, with a unique solution. Both methods optimize the marginal log-likelihood $\log p(Y_\Omega \mid W_r, \lambda)$, which is identified up to a right-orthogonal rotation $W_r \mapsto W_r R$; this rotation ambiguity creates saddle points but no spurious local maxima, as all critical points satisfying $\partial \mathcal{L} / \partial W_r = 0$ yield the same posterior mean $\mu_i = W_r \hat{z}_i$ and marginal covariance $W_r A_i^{-1} W_r^\top$.

5 Implementation

A reference implementation is provided in the `matlap` Python package using JAX [6] for JIT compilation and automatic differentiation.

Numerical stability. Matrix inverses are never formed explicitly. The matrix square root $Q = \Psi^{1/2}$ is obtained via `jnp.linalg.eigh`, which returns eigenvalues d_k and eigenvectors V so that $Q^{-1} = V \text{diag}(1/d_k) V^\top$ can be applied matrix-free. The per-row precision system $(D_i + \bar{\lambda}Q^{-1})\mu_i = D_i y_i$ is solved with a Cholesky factorisation (`jax.scipy.linalg.cho_factor/cho_solve`).

Vectorisation. The m row updates are independent and are vectorised with `jax.vmap`, giving a single batched Cholesky solve per iteration.

ELBO monotonicity. After updating all rows, Ψ and Q are recomputed from the new $\{\mu_i, \Sigma_i\}$ before the ELBO is evaluated. This ensures the ELBO is computed at a fully self-consistent state and is guaranteed non-decreasing.

Lambda grid search. As an alternative to the empirical-Bayes update, `matlap_grid` accepts a user-supplied grid of λ values, runs CAVI with λ held fixed at each grid point, and returns the solution with the highest ELBO.

6 Scalable Extensions

6.1 Randomized SVD for Approximate Nuclear Norm in SVI

For the gradient-based SVI comparators, the bottleneck is computing $\|X\|_* = \sum_i \sigma_i(X)$ via full SVD at each gradient step, costing $O(mn \min(m, n))$. We replace this with a rank- r randomized SVD approximation:

$$\|X\|_* \approx \sum_{k=1}^r \hat{\sigma}_k(X), \quad (35)$$

where $\hat{\sigma}_1 \geq \dots \geq \hat{\sigma}_r$ are the top- r singular values estimated by power-iteration randomized SVD [7]. The method forms a sketch $Y_\Omega = (XX^\top)^q X\Omega$ for a Gaussian random matrix $\Omega \in \mathbb{R}^{n \times (r+p)}$ with p oversampling columns, then extracts approximate singular values from the resulting small matrix.

To enable gradient-based optimization, we implement the approximation with a `jax.custom_vjp` rule. The forward pass returns $\sum_k \hat{\sigma}_k$; the backward pass propagates the gradient $g \mapsto g \hat{U}_r \hat{V}_r^\top$, which is the standard nuclear norm subgradient restricted to the top- r component and satisfies $\|\nabla\|_{\text{op}} \leq 1$. This reduces the per-step cost from $O(mn^2)$ to $O(mnr)$ for matrices with $n \leq m$.

6.2 Low-Rank Grid Search with Warm-Started Regularisation Path

Rather than estimating λ jointly with X via the empirical-Bayes update, one may instead search a discrete grid of candidate values and select the best by ELBO or another post-hoc score, as in `matlap_grid`. We combine this with the low-rank CAVI (Section 3.2) to obtain `matlap_grid_lowrank`, which is both scalable and avoids the misspecification of the automatic λ in factor space.

The grid is traversed in *decreasing* order of λ (strong to weak regularisation). The solution at the previous (larger) λ warm-starts the next grid point: the shared loading matrix V_r and per-row factor means $\hat{z}_i$ are reused as the initial state. Because the previous solution is already near the new optimum (the signal grows smoothly as λ decreases), each grid point typically converges in far fewer CAVI iterations than a cold start. After all grid points are evaluated, the λ yielding the highest ELBO (computed in factor space, consistent with Section 3.2) is selected.

The same warm-started path is also used by `matlap_grid_lowrank_isotropic`, which retains full n -dimensional entropy and can score λ by ELBO, closed-form leave-one-out, Rényi α -ELBO, or the $\alpha = 0$ importance-style objective. In the lambda-selection study below, the Rényi score tracked cross-validation most closely, while the LOO score was consistently too conservative.

This strategy costs $O(G \cdot mn r)$ in total, where G is the number of grid points. On a GPU at $m = 10,000$, $n = 1,000$, $r = 30$, a grid of $G = 7$ points converges in under one second.

6.3 Scalable Variational Guides for SVI

Two additional guide families reduce the SVI memory footprint to $O(mn)$ while capturing structured posterior correlations:

Shared column-factor guide (matrix_factor). A single factor matrix $F \in \mathbb{R}^{n \times k}$ is shared across all m rows:

$$q(x_i) = \mathcal{N}(x_i \mid \mu_i, FF^\top + \text{diag}(\psi_i)), \quad (36)$$

where $\psi_i \in \mathbb{R}^n$ is a per-row diagonal. This captures shared column-space correlations (as in matrix normal approximations) at $O(mn + nk)$ parameter cost—the same asymptotic order as a fully-factorised guide.

Per-row low-rank guide (row_lowrank). Each row i has its own low-rank factor $F_i \in \mathbb{R}^{n \times k}$:

$$q(x_i) = \mathcal{N}(x_i \mid \mu_i, F_i F_i^\top + \text{diag}(\psi_i)). \quad (37)$$

At rank $k = 5$ this costs approximately 600 MB for $10,000 \times 1,000$, versus 40 GB for full row-MVN covariances.

Both guides are implemented using `numpyro`'s `LowRankMultivariateNormal` distribution and combined with the rSVD nuclear norm approximation, reducing per-step cost to $O(mnr)$.

6.4 Factor Analysis EM and Gradient Marginal Likelihood

These are comparator methods. They treat the loading matrix $W_r \in \mathbb{R}^{n \times r}$ as an unconstrained free parameter and optimise the penalised marginal log-likelihood.

Model. We write $x_i = W_r z_i$ with isotropic prior $z_i \sim \mathcal{N}(0, \lambda^{-1} I_r)$ and Gaussian likelihood $y_{ij} \mid x_i \sim \mathcal{N}(x_{ij}, s_{ij}^2)$ for observed entries. Unlike the low-rank CAVI, which fixes $\text{col}(W_r)$ and writes $W_r = V_r \text{diag}(d_r)$, here W_r is unconstrained and absorbs both directions and scales.

Marginal log-likelihood. Integrating out z_i via the Woodbury identity gives, for each row i ,

$$\log p(y_i^\Omega \mid W_r, \lambda) = -\frac{1}{2} \left[y_i^\top \Lambda_i y_i - \text{rhs}_i^\top A_i^{-1} \text{rhs}_i - \sum_{j \in \Omega_i} \log p_{ij} + \log |A_i| - r \log \lambda + |\Omega_i| \log 2\pi \right], \quad (38)$$

where $\Lambda_i = \text{diag}(p_i)$, $A_i = \lambda I_r + W_r^\top \Lambda_i W_r \in \mathbb{R}^{r \times r}$, and $\text{rhs}_i = W_r^\top \Lambda_i y_i$. Only the $r \times r$ matrix A_i need be inverted per row.

E-step. The posterior over z_i is $q(z_i) = \mathcal{N}(\hat{z}_i, A_i^{-1})$ with

$$A_i = \lambda I_r + W_r^\top \Lambda_i W_r, \quad \hat{z}_i = A_i^{-1} W_r^\top (p_i \odot y_i). \quad (39)$$

The posterior second moment is $\mathbb{E}[z_i z_i^\top] = A_i^{-1} + \hat{z}_i \hat{z}_i^\top$ and the posterior mean of x_i is $\mu_i = W_r \hat{z}_i$.

FA EM M-step. The expected complete-data log-likelihood decouples across rows of $W_r^\top$. Setting the gradient for the j -th row $w_j^\top$ to zero yields the closed-form update

$$w_j = \left(\sum_i p_{ij} \mathbb{E}[z_i z_i^\top] \right)^{-1} \sum_i p_{ij} y_{ij} \hat{z}_i, \quad (40)$$

a weighted least-squares problem with a unique solution. The M-step for λ is the Gamma posterior mean,

$$\bar{\lambda} = \frac{a_0 + mr/2}{b_0 + \frac{1}{2} \sum_i \text{Tr}(\mathbb{E}[z_i z_i^\top])}. \quad (41)$$

Per-iteration cost is $O(mnr + mr^3)$, identical to the low-rank CAVI.

Gradient marginal likelihood. As an alternative to EM, we directly maximise

$$\mathcal{L}_{\text{grad}}(W_r, \lambda) = \sum_i \log p(y_i^\Omega \mid W_r, \lambda) + (a_0 - 1) \log \lambda - b_0 \lambda \quad (42)$$

with respect to W_r and $\log \lambda$ using the Adam optimiser [8], with gradients computed by JAX autodiff through (38).

7 Empirical Comparison

We benchmark all scalable methods on a synthetic matrix completion problem with $m = 10,000$ rows, $n = 1,000$ columns, true rank 15, heteroscedastic Gaussian noise ($\sigma_{ij} \sim \text{Uniform}(0.5, 1.5)$), and 20% of entries held out as a test set. We report test-set RMSE and wall-clock time on a NVIDIA RTX 3090 GPU, averaged over 3 seeds.

Methods. `proximal` and `proximal_cv` solve the nuclear-norm penalised least-squares problem via FISTA (300 iterations); λ is set by a heuristic or 5-fold entry-wise CV respectively. `matlap_lowrank` runs the rank-30 low-rank CAVI with automatic empirical-Bayes λ (80 iterations, converged). `matlap_grid_lowrank` runs the same low-rank CAVI on a 7-point log-spaced λ grid with warm-started path (converged at each point). `matlap_grid_lowrank_isotropic` runs the low-rank-plus-isotropic CAVI path and can score λ by ELBO, LOO, or Rényi α -ELBO ($\alpha = 0$ is the importance-style limit). `matlap_batched` runs exact full CAVI with batch size 64, automatic λ (converged; $O(n^3)$ per row, so slow at $n = 1,000$). `vi_diagonal` runs SVI with a fully-factorised Gaussian guide and full SVD nuclear norm (1000 steps, not converged). `vi_diagonal_approx`, `vi_matrix_factor`, and `vi_row_lowrank` use the rSVD nuclear norm approximation (rank 30) with structured guides (1000 steps, not converged).

Results.

Discussion. `matlap_grid_lowrank` achieves the lowest RMSE (0.092) and the shortest runtime (0.6 s), demonstrating that the warm-started regularisation path is both accurate and highly efficient. By evaluating multiple λ values on a shared low-rank basis, it avoids the over-shrinkage of the automatic empirical-Bayes λ in factor space (which inflates the effective regularisation by a factor of $n/r \approx 33$) while remaining far cheaper than the proximal CV baseline.

Table 1: Test-set RMSE and GPU runtime on a $10,000 \times 1,000$ rank-15 matrix, 3 seeds.

Method	RMSE	Time (s)	Converged	Notes
matlap_grid_lowrank	0.092	0.6	✓	Best overall
proximal_cv	0.105	346	–	$500\times$ slower
proximal	0.123	67	–	Fixed λ
matlap_batched	0.153	185	✓	Exact; slow at $n = 1,000$
matlap_lowrank	0.257	1.7	✓	Auto- λ over-shrinks
vi_diagonal	0.207	203	–	1000 steps
vi_diagonal_approx	0.654	52	–	rSVD approx unstable
vi_matrix_factor	0.530	93	–	rSVD approx unstable
vi_row_lowrank	0.529	141	–	rSVD approx unstable

matlap_batched converges to the exact full-CAVI solution (RMSE 0.153) but is slow at $n = 1,000$ because each row requires an $O(n^3)$ Cholesky; it is intended for settings with moderate n ($\lesssim 300$) where exact row posteriors are needed without the full $O(mn^2)$ memory.

The SVI methods (vi_diagonal, vi_matrix_factor, vi_row_lowrank) have not converged at 1000 steps, and the rSVD nuclear-norm approximation introduces additional gradient noise at rank 30 that degrades the approx variants. With sufficient steps and a higher approximation rank, vi_diagonal achieves competitive RMSE (0.207 at 1000 steps).

7.1 Lambda Selection Study

We also compared λ -selection rules on a 500×100 synthetic problem under two sweeps: varying true rank (1, 3, 5, 10, 20, 40) and varying SNR (0.1, 0.3, 1, 3, 10), with 3 seeds per setting. Table 2 reports mean test RMSE averaged over each sweep.

Table 2: Mean test RMSE across the rank and SNR sweeps.

Method	Rank sweep	SNR sweep
proximal_cv	0.467	0.314
matlap_grid	0.615	0.450
lowrank_cv	0.600	0.485
lowrank_grid	0.637	0.634
iso_cv	0.585	0.445
iso_grid_renyi	0.585	0.446
iso_grid_is	0.587	0.464
iso_grid_loo	0.631	0.476

The key pattern is that iso_grid_renyi is essentially tied with iso_cv across both sweeps, while iso_grid_is ($\alpha = 0$) is close but degrades at higher SNR and iso_grid_loo is too conservative. The ELBO-based grids (matlap_grid and lowrank_grid) remain systematically worse than CV.

References

- [1] Daniel D. Lee and H. Sebastian Seung. “Learning the parts of objects by non-negative matrix factorization”. In: *Nature* 401.6755 (1999), pp. 788–791.

- [2] T. Cai, X. Li, and Z. Ma. “Optimal rates of convergence for noisy sparse phase retrieval via thresholded Wirtinger flow”. In: *Annals of Statistics* 44.5 (2021), pp. 2221–2251.
- [3] Benjamin Recht, Maryam Fazel, and Pablo A. Parrilo. “Guaranteed minimum-rank solutions of linear matrix equations via nuclear norm minimization”. In: *SIAM Review* 52.3 (2010), pp. 471–501.
- [4] Emmanuel J. Candès and Benjamin Recht. “Exact matrix completion via convex optimization”. In: *Communications of the ACM* 55.6 (2012), pp. 111–119.
- [5] Yingzhen Li and Richard E. Turner. “Rényi divergence variational inference”. In: *Advances in Neural Information Processing Systems (NeurIPS)*. 2016.
- [6] James Bradbury et al. *JAX: composable transformations of Python+NumPy programs*. <http://github.com/google/jax>. 2018.
- [7] Nathan Halko, Per-Gunnar Martinsson, and Joel A. Tropp. “Finding structure with randomness: Probabilistic algorithms for constructing approximate matrix decompositions”. In: *SIAM Review* 53.2 (2011), pp. 217–288.
- [8] Diederik P. Kingma and Jimmy Ba. “Adam: A method for stochastic optimization”. In: *International Conference on Learning Representations (ICLR)*. 2015.